

5.11.2016 (E)

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Exam
Nov – Dec 2016

Max. Marks: 100

Class: T. Y. B. Tech.

Name of the Course: Operating System

Course Code: UCEC501

Duration: 3 Hours

Semester: V

Branch: Computer

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Question No.		Max. Marks
Q. 1	Attempt any Two questions out of remaining three questions.	
(a)	Define Operating System. Describe the following types of operating system a) Serial Processing b) Simple Batch system c) Multiprogrammed Batch System d) Time-Sharing System	10
(b)	Write a Shell script to find factorial of a given integer.	10
(c)	Explain various types of files system of UNIX. List the system calls for Unix file system.	10
Q. 2 (a)	What is a thread? Explain the various types of threads in detail.	10
Q. 2 (b)	Draw a five state process model diagram and explain the various states that a process passes through in its lifetime.	10
Q. 3 (a)	Define the bounded buffer producer consumer problem and give a solution for the same using semaphore.	10
Q. 3 (b)	What is deadlock? Describe a resource allocation graph with example, i) with a dead lock ii) with a cycle but no dead lock OR Explain the deadlock prevention and integrated deadlock strategy.	10

Q. 4 (a)	List and describe the mutual exclusion hardware support algorithm. OR <i>note</i> Write a short on message passing.	10																				
Q. 4 (b)	Consider the following set of processes. <table border="1"><thead><tr><th>Process</th><th>P1</th><th>P2</th><th>P3</th><th>P4</th></tr></thead><tbody><tr><td>Arrival Time</td><td>0</td><td>2</td><td>3</td><td>5</td></tr><tr><td>Burst Time</td><td>10</td><td>5</td><td>2</td><td>20</td></tr><tr><td>Priority</td><td>2</td><td>1</td><td>0</td><td>3</td></tr></tbody></table> Draw Gantt Charts and calculate average waiting time, average turnaround time using following CPU scheduling algorithms. i. First Come First Serve (FCFS) ii. Preemptive Shortest Job First (SJF) (0=High Priority). iii. Non preemptive priority (0=High Priority)	Process	P1	P2	P3	P4	Arrival Time	0	2	3	5	Burst Time	10	5	2	20	Priority	2	1	0	3	10
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Q5 (a)	List and Describe the various approaches for I/O Buffering provided by OS. OR List the criteria in choosing a file organization. Describe advantage and disadvantage of the following file organization techniques a. Pile b. Sequential File c. Indexed Sequential File	10																				
Q5 (b)	Suppose that a disk has 200 tracks, numbers 0 to 199. The read/write head is currently serving at track 100. The queue of pending requests is in order: 55, 58, 39, 18, 90, 160, 150, 38, and 184. For each disk scheduling algorithms: FCFS, SSTF and C-SCAN. a) Show the graphical representation for above scheduling algorithms. b) Find the average head movement for above scheduling algorithms	10																				